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apple_retail_sales_analytics_using_postgreSQL.sql Processes

apple_db/postgres@PostgreSQL 18

Query Query History

```
1 -- Apple retails Millions Rows Sales Schemas
2
3 -- DROP Table Command
4 DROP TABLE IF EXISTS warranty;
5 DROP TABLE IF EXISTS sales;
6 DROP TABLE IF EXISTS products;
7 DROP TABLE IF EXISTS category; -- parent
8 DROP TABLE IF EXISTS stores; -- parent
9
10 -- CREATE TABLE COMMANDS
11
12 CREATE TABLE stores(
13   store_id VARCHAR(5) PRIMARY KEY,
14   store_name VARCHAR(30),
15   city VARCHAR(25),
16   country VARCHAR(25)
17 );
18
19 DROP TABLE IF EXISTS category;
20 CREATE TABLE category(
21   category_id VARCHAR(10) PRIMARY KEY,
22   category_name VARCHAR(20)
23 );
24
25 CREATE TABLE products(
26   product_id VARCHAR(10) PRIMARY KEY,
27   product_name VARCHAR(35),
28   category_id VARCHAR(10),
29   launch_date date,
30   price FLOAT,
31   CONSTRAINT fk_category FOREIGN KEY (category_id) REFERENCES category(category_id)
```

Total rows: 326 Query complete 00:00:00.828

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7 DROP TABLE IF EXISTS category; -- parent
8 DROP TABLE IF EXISTS stores; -- parent
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10 -- CREATE TABLE COMMANDS
11
12 CREATE TABLE stores(
13   store_id VARCHAR(5) PRIMARY KEY,
14   store_name VARCHAR(30),
15   city VARCHAR(25),
16   country VARCHAR(25)
17 );
18
19 DROP TABLE IF EXISTS category;
20 CREATE TABLE category(
21   category_id VARCHAR(10) PRIMARY KEY,
22   category_name VARCHAR(20)
23 );
24
25 CREATE TABLE products(
26   product_id VARCHAR(10) PRIMARY KEY,
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Total rows: 326 Query complete 00:00:00.828

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```
25 CREATE TABLE products(  
26 product_id VARCHAR(10) PRIMARY KEY,  
27 product_name VARCHAR(35),  
28 category_id VARCHAR(10),  
29 launch_date date,  
30 price FLOAT,  
31 CONSTRAINT fk_category FOREIGN KEY (category_id) REFERENCES category(category_id)  
32 );  
33  
34 CREATE TABLE sales(  
35 sale_id VARCHAR(15) PRIMARY KEY,  
36 sale_date DATE,  
37 store_id VARCHAR(10), -- this fk  
38 product_id VARCHAR(10), -- this fk  
39 quantity INT,  
40 CONSTRAINT fk_store FOREIGN KEY (store_id) REFERENCES stores(store_id),  
41 CONSTRAINT fk_product FOREIGN KEY (product_id) REFERENCES products(product_id)  
42 );  
43 DROP TABLE IF EXISTS warranty;  
44 CREATE TABLE warranty(  
45 claim_id VARCHAR(10) PRIMARY KEY,  
46 claim_date date,  
47 sale_id VARCHAR(15),  
48 repair_status VARCHAR(15),  
49 CONSTRAINT fk_sales FOREIGN KEY (sale_id) REFERENCES sales(sale_id)  
50 );  
51  
52 -- Success Message  
53  
54 SELECT 'Schema created successful' as Success_Message;  
55
```

Total rows: 326
Query complete 00:00:00.828

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Query History
136
137 -- 1. Find the number of stores in each country.
138
139 SELECT
140 country,
141 count(store_id) as count_stores
142 FROM stores
143 GROUP BY country
144 ORDER BY 2 DESC;
145
146 -- 2. Calculate the total number of units sold by each store.
147
148 -- v1
149 SELECT
150 store_id,
151 sum(quantity) as total_unit_sold
152 FROM sales
153 GROUP BY store_id;
154
155 -- v2
156 SELECT
157 s.store_id,
158 st.store_name,
159 sum(s.quantity) as total_unit_sold
160 FROM sales as s
161 JOIN
162 stores as st
163 ON st.store_id = s.store_id
164 GROUP BY 1, 2
165 ORDER BY 3 DESC
166
Total rows: 326
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Query History
165 ORDER BY 3 DESC
166
167 -- 3. Identify how many sales occurred in December 2023.
168
169 SELECT
170 *
171 --TO_CHAR(sale_date, 'MM-YYYY')
172 FROM sales
173 WHERE TO_CHAR(sale_date, 'MM-YYYY') = '12-2023';
174 --
175 SELECT
176 count(sale_id) as total_sales
177 FROM sales
178 WHERE TO_CHAR(sale_date, 'MM-YYYY') = '12-2023';
179
180
181 -- 4. Determine how many stores have never had a warranty claim filed.
182
183 SELECT count(*) FROM stores
184 WHERE store_id NOT IN (
185 SELECT
186 DISTINCT store_id
187 FROM sales as s
188 RIGHT JOIN warranty as w
189 ON s.sale_id = w.sale_id
190);
191
192
193

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202
203 -- 5. Calculate the percentage of warranty claims marked as "In Progress".
204
205 SELECT * from warranty;
206
207 SELECT DISTINCT repair_status from warranty;
208
209 SELECT
210 ROUND
211 (COUNT(claim_id)/
212 (SELECT COUNT(*) FROM warranty)::numeric
213 * 100
214 ,2) as warranty_in_progress
215 FROM warranty
216 WHERE repair_status = 'In Progress';
217
218 -- 6. Identify which store had the highest total units sold in the last year.
219
220
221 SELECT
222 s.store_id,
223 st.store_name,
224 sum(s.quantity)
225 FROM sales as s
226 JOIN stores as st
227 ON s.store_id = st.store_id
228 WHERE sale_date >= (CURRENT_DATE - INTERVAL '3 year')
229 GROUP BY 1, 2
230 ORDER BY 2 DESC
231 LIMIT 1
232
Total rows: 326 Query complete 00:00:00.828 CRLF Ln 219, Col 1

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241
242 -- 7. Count the number of unique products sold in the last year.
243
244 select * from products
245 select * from sales
246
247 SELECT
248 DISTINCT s.product_id,
249 p.product_name
250 FROM sales as s
251 JOIN products as p
252 ON s.product_id = p.product_id
253 WHERE sale_date >= (CURRENT_DATE - INTERVAL '3 year')
254 GROUP BY 1,2;
255
256 SELECT
257 COUNT(DISTINCT product_id)
258 FROM sales
259 WHERE sale_date >= (CURRENT_DATE - INTERVAL '3 year')
260
261 -- 8. Find the average price of products in each category.
262
263 SELECT
264 p.category_id,
265 c.category_name,
266 avg(price) as avg_price
267 FROM products as p
268 JOIN category as c
269 ON p.category_id = c.category_id
270 GROUP BY 1, 2
271 ORDER BY 3 DESC
272
Total rows: 326 Query complete 00:00:00.828
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Query
Query History
272
273 -- 9. How many warranty claims were filed in 2020?
274
275 SELECT
276 COUNT(*) as warranty_claim
277 FROM warranty
278 WHERE EXTRACT(YEAR FROM claim_date) = 2024
279
280 -- 10. For each store, identify the best-selling day based on highest quantity sold.
281
282 SELECT *
283 FROM(
284 SELECT
285 store_id,
286 TO_CHAR(sale_date, 'Day') as day_name,
287 SUM(quantity) as total_unit_sold,
288 RANK() OVER(PARTITION BY store_id ORDER BY SUM(quantity) DESC) as rank
289 FROM sales
290 GROUP BY 1, 2
291) as t1
292 WHERE rank = 1
293
294
295
296
297
298
299
300
301
302
Total rows: 326 Query complete 00:00:00.828 CRLF Ln 301, Col 1

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Query History
-- 11. Identify the least selling product in each country for each year based on total units sold.
WITH product_rank
AS
(
SELECT
st.country,
p.product_name,
sum(s.quantity) as total_gty_sold,
RANK() OVER(PARTITION BY st.country ORDER BY SUM(s.quantity)) as rank
FROM sales as s
JOIN
stores as st
ON s.store_id = st.store_id
JOIN
products as p
ON s.product_id = p.product_id
GROUP BY 1, 2
)

SELECT
*
FROM product_rank
WHERE rank = 1

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328
329 -- 12. Calculate how many warranty claims were filed within 180 days of a product sale.
330
331 SELECT
332 w.*,
333 s.sale_date,
334 w.claim_date - s.sale_date
335 FROM warranty as w
336 LEFT JOIN
337 sales as s
338 ON s.sale_id = w.sale_id
339 WHERE
340 w.claim_date - s.sale_date <= 180
341
342
343 SELECT
344 count(*)
345 FROM warranty as w
346 LEFT JOIN
347 sales as s
348 ON s.sale_id = w.sale_id
349 WHERE
350 w.claim_date - s.sale_date <= 180
351

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352 -- 13. Determine how many warranty claims were filed for products launched in the last two years.
353
354 SELECT
355 p.product_name,
356 COUNT(w.claim_id) as no_claim,
357 COUNT(s.sale_id)
358 FROM warranty as w
359 RIGHT JOIN
360 sales as s
361 ON s.sale_id = w.sale_id
362 JOIN products as p
363 ON p.product_id = s.product_id
364 WHERE p.launch_date >= CURRENT_DATE - INTERVAL '3 years'
365 GROUP BY 1
366 HAVING COUNT(w.claim_id) > 0
367
368 -- 14. List the months in the last three years where sales exceeded 5,000 units in the USA.
369
370 SELECT
371 TO_CHAR(sale_date, 'MM-YYYY') as month,
372 SUM(s.quantity) as total_unit_sold
373 FROM sales as s
374 JOIN
375 stores as st
376 ON s.store_id = st.store_id
377 WHERE
378 st.country = 'USA'
379 AND
380 s.sale_date >= CURRENT_DATE - INTERVAL '3 year'
381 GROUP BY 1
382 HAVING SUM(s.quantity)> 5000
Total rows: 326 Query complete 00:00:00.828
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380 s.sale_date >= CURRENT_DATE - INTERVAL '3 year'
381 GROUP BY 1
382 HAVING SUM(s.quantity)> 5000
383
384 -- 15. Identify the product category with the most warranty claims filed in the last two years.
385
386 SELECT
387 c.category_name,
388 COUNT(w.claim_id) as total_claims
389 FROM warranty as w
390 LEFT JOIN
391 sales as s
392 ON w.sale_id = s.sale_id
393 JOIN products as p
394 ON p.product_id = s.product_id
395 JOIN
396 category as c
397 ON c.category_id = p.category_id
398 WHERE
399 w.claim_date >= CURRENT_DATE - INTERVAL '2 year'
400 GROUP BY 1
401
402
403
404
405
406
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410
Total rows: 326 Query complete 00:00:00.828 CRLF Ln 409, Col 1

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-- 16. Determine the percentage chance of receiving warranty claims after each purchase for each country.
SELECT
country,
total_unit_sold,
total_claim,
COALESCE(total_claim::numeric/total_unit_sold:: numeric * 100, 0)
as risk
FROM
(SELECT
st.country,
SUM(s.quantity) as total_unit_sold,
COUNT(w.claim_id) as total_claim
FROM sales as s
JOIN stores as st
ON s.store_id = st.store_id
LEFT JOIN
warranty as w
ON w.sale_id = s.sale_id
GROUP BY 1) t1
ORDER BY 4 DESC

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435 -- 17. Analyze the year-by-year growth ratio for each store.
436
437 WITH yearly_sales
438 AS
439 (
440 SELECT
441 s.store_id,
442 st.store_name,
443 EXTRACT(YEAR FROM sale_date) as year,
444 SUM(s.quantity * p.price) as total_sale
445 FROM sales as s
446 JOIN
447 products as p
448 ON s.product_id = p.product_id
449 JOIN stores as st
450 ON st.store_id = s.store_id
451 GROUP BY 1, 2, 3
452 ORDER BY 2, 3
453),
454 growth_ratio
455 AS
456 (
457 SELECT
458 store_name,
459 year,
460 LAG(total_sale, 1) OVER(PARTITION BY store_name ORDER BY year) as last_year_sale,
461 total_sale as current_year_sale
462 FROM yearly_sales
463)
464
465 SELECT

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449 JOIN stores as st
450 ON st.store_id = s.store_id
451 GROUP BY 1, 2, 3
452 ORDER BY 2, 3
453),
454 growth_ratio
455 AS
456 (
457 SELECT
458 store_name,
459 year,
460 LAG(total_sale, 1) OVER(PARTITION BY store_name ORDER BY year) as last_year_sale,
461 total_sale as current_year_sale
462 FROM yearly_sales
463)
464
465 SELECT
466 store_name,
467 year,
468 last_year_sale,
469 current_year_sale,
470 ROUND(
471 (current_year_sale - last_year_sale):: numeric/
472 last_year_sale::numeric * 100
473 ,3) as growth_ratio
474 FROM growth_ratio
475 WHERE
476 last_year_sale IS NOT NULL
477 AND
478 YEAR <> EXTRACT(YEAR FROM CURRENT_DATE)
479
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-- 18. Calculate the correlation between product price and warranty claims for products sold in the last five years, segmented by price range.

SELECT
 CASE
 WHEN p.price < 500 THEN 'Less Expenses Product'
 When p.price BETWEEN 500 AND 1000 THEN 'Mid range Product'
 ELSE 'Expensive Product'
 END as price_segment,
 COUNT(w.claim_id) as total_Claim
FROM warranty as w
LEFT JOIN
sales as s
ON w.sale_id = s.sale_id
JOIN
products as p
ON p.product_id = s.product_id
WHERE claim_date >= CURRENT_DATE - INTERVAL '5 year'
GROUP BY 1

-- 19. Identify the store with the highest percentage of "Paid Repaired" claims relative to total claims filed.

SELECT DISTINCT repair_status from warranty

WITH paid_repair
AS
(SELECT
 s.store_id,
 COUNT(w.claim_id) as paid_Repaired
FROM sales as s

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510 COUNT(w.claim_id) as paid_repaired
511 FROM sales as s
512 RIGHT JOIN warranty as w
513 ON w.sale_id = s.sale_id
514 WHERE w.repair_status = 'Paid Repaired'
515 GROUP BY 1
516),
517
518 total_repaired
519 AS
520 (SELECT
521 s.store_id,
522 COUNT(w.claim_id) as total_repaired
523 FROM sales as s
524 RIGHT JOIN warranty as w
525 ON w.sale_id = s.sale_id
526 GROUP BY 1)
527
528 SELECT
529 tr.store_id,
530 st.store_name,
531 pr.paid_repaired,
532 ROUND(pr.paid_repaired::numeric/
533 tr.total_repaired::numeric * 100
534 ,2) as percentage_paid_repaired
535 FROM paid_repair as pr
536 JOIN
537 total_repaired tr
538 ON pr.store_id = tr.store_id
539 JOIN stores as st
540 ON tr.store_id = st.store_id
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543 -- 20. Write a query to calculate the monthly running total of sales for each store over the past four years
544 -- and compare trends during this period.
545
546
547 WITH monthly_sales
548 AS
549 (SELECT
550 store_id,
551 EXTRACT(YEAR FROM sale_date) as year,
552 EXTRACT(MONTH FROM sale_date) as month,
553 SUM(p.price * s.quantity) as total_revenue
554 FROM sales as s
555 JOIN
556 products as p
557 ON s.product_id = p.product_id
558 GROUP BY 1, 2, 3
559 ORDER BY 1, 2, 3
560)
561
562 SELECT
563 store_id,
564 month,
565 year,
566 total_revenue,
567 SUM(total_revenue) OVER(PARTITION BY store_id ORDER BY year, month) as running_total
568 FROM monthly_sales
569
570
571 -- Bonus Question
572
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570
571 -- Bonus Question
572
573 -- 21. Analyze product sales trends over time, segmented into key periods: from launch to 6 months, 6-12 months, 12-18 months, and beyond 18 mont
574
575 SELECT
576     p.product_name,
577     CASE
578         WHEN s.sale_date BETWEEN p.launch_date AND p.launch_date + INTERVAL '6 month' THEN '0-6 month'
579         WHEN s.sale_date BETWEEN p.launch_date + INTERVAL '6 month' AND p.launch_date + INTERVAL '12 month' THEN '6-12'
580         WHEN s.sale_date BETWEEN p.launch_date + INTERVAL '12 month' AND p.launch_date + INTERVAL '18 month' THEN '12-18'
581         ELSE '18 +'
582     END as plc,
583     SUM(s.quantity) as total_qty_sale
584
585 FROM sales as s
586 JOIN products as p
587 ON s.product_id = p.product_id
588 GROUP BY 1, 2
589 ORDER BY 1, 3 DESC
590
591
592
593
594
595
596
597
598
```

Total rows: 326
Query complete 00:00:00.828
CRLF
Ln 598, Col 1